

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 14

Assign AdministratorAccess Permission Set to the Administrators SSO Group

IAM Identity Center | Permission Sets | Account Assignments

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 13, 2026
STATUS	Complete

Executive Summary

Created an **AdministratorAccess** permission set in AWS IAM Identity Center, assigned that permission set to the **Administrators** group across two AWS accounts, and confirmed that the SSO user in the Administrators group could see the assigned accounts through the AWS access portal. This project completes the first usable SSO administrator access path for the portfolio environment.

Business Problem

Project 13 created the IAM Identity Center identity foundation, but the SSO user still needed authorization into AWS accounts. In IAM Identity Center, access is granted by mapping a user or group to a permission set and one or more AWS accounts. Using a group assignment lets future administrative users inherit the same access pattern without one-off account permissions.

Objectives

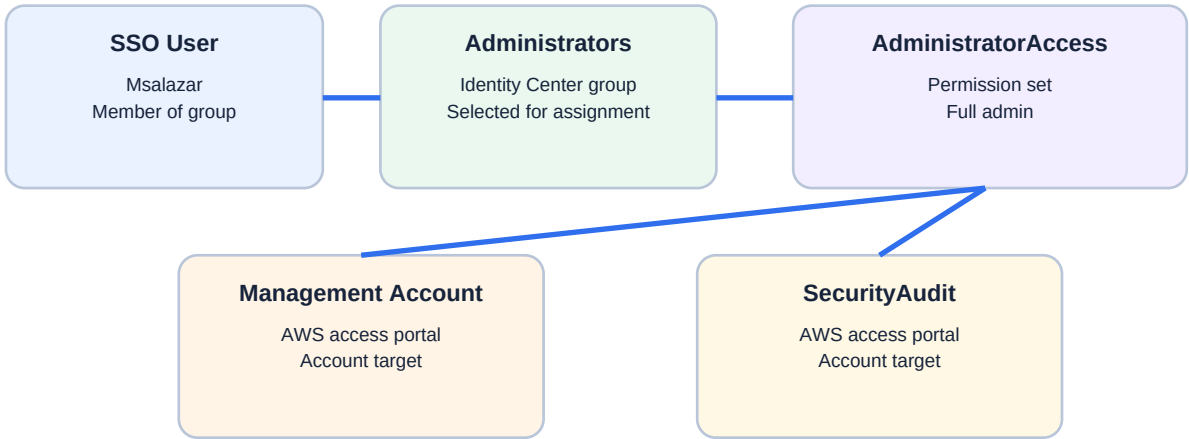
- Create an AdministratorAccess permission set.
- Use the predefined AdministratorAccess managed policy pattern.
- Assign the Administrators group to the permission set.
- Apply the assignment to two AWS accounts in the organization.
- Allow the SSO user in Administrators to access assigned accounts from the AWS access portal.
- Document the distinction between SSO identity, permission sets, and account assignments.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	IAM Identity Center
SSO User	Msalazar / Michael Salazar
Identity Center Group	Administrators
Permission Set	AdministratorAccess
Permission Set Description	Full admin
Assignment Targets	Two AWS accounts
Visible Access Portal Accounts	Management account and SecurityAudit account
Session Duration	Guide default: 1 hour; provided evidence does not independently show final duration
Access Model	Group + permission set + account

Evidence Limitation	Evidence confirms assignment and portal visibility; no separate screenshot of a privileged action executed inside an account was provided.
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Architecture



Identity Center grants account access by mapping group + permission set + AWS account.
AWS provisions reserved SSO roles in target accounts behind the scenes.

Figure 1 - IAM Identity Center account assignment model for administrator access.

Implementation Summary

1. Opened IAM Identity Center from the management account.
2. Created a new permission set named AdministratorAccess.
3. Used the predefined AdministratorAccess permission set option with a Full admin description.
4. Selected two AWS accounts for assignment.
5. Selected the Administrators Identity Center group.
6. Selected the AdministratorAccess permission set.
7. Submitted the assignment so IAM Identity Center could provision access into the selected accounts.
8. Confirmed that the SSO access portal showed two AWS accounts available to the user in the Administrators group.

KEY CONCEPT

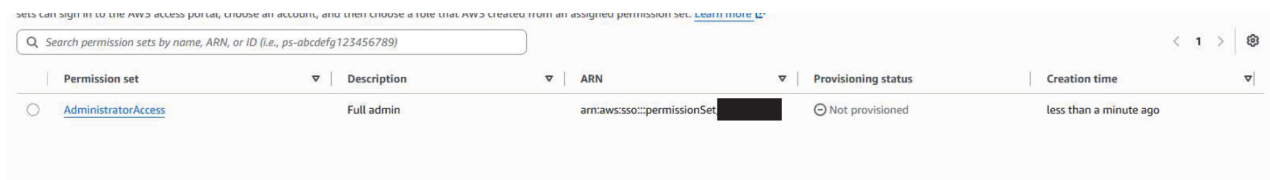
IAM Identity Center access is not granted by group membership alone. The effective access comes from the three-part mapping: **group + permission set + account**.

Authorization Model

SSO User	The workforce identity that signs in through the AWS access portal.
Administrators Group	The Identity Center group used to manage admin access as a role-like collection.
Permission Set	The policy package named AdministratorAccess that defines what the user can do.
AWS Accounts	The management account and SecurityAudit account that receive the assignment.
Provisioned Role	IAM Identity Center provisions AWSReservedSSO roles in target accounts behind the scenes.

Evidence

Figure 2 - AdministratorAccess Permission Set Created



The screenshot shows the AWS IAM console with a search bar at the top. Below the search bar, a table lists the permission sets. The first row is for the 'AdministratorAccess' permission set, which has a description of 'Full admin', an ARN of 'arn:aws:sso::permissionSet:...', a provisioning status of 'Not provisioned', and a creation time of 'less than a minute ago'.

Permission set	Description	ARN	Provisioning status	Creation time
AdministratorAccess	Full admin	arn:aws:sso::permissionSet:...	Not provisioned	less than a minute ago

The redacted IAM Identity Center permission set evidence shows a permission set named AdministratorAccess with description Full admin. The screenshot was captured immediately after creation while provisioning status still showed Not provisioned.

Figure 3 - Administrators Group Assigned AdministratorAccess to Two AWS Accounts

Review and submit

Review and submit assignments to 2 AWS accounts

Step 1: Select users and groups

Users and groups (1)

Username / group name ↗

[Administrators](#) ↗

Group

Step 2: Select permission sets

Permission sets (1)

Permission set

Description

ARN

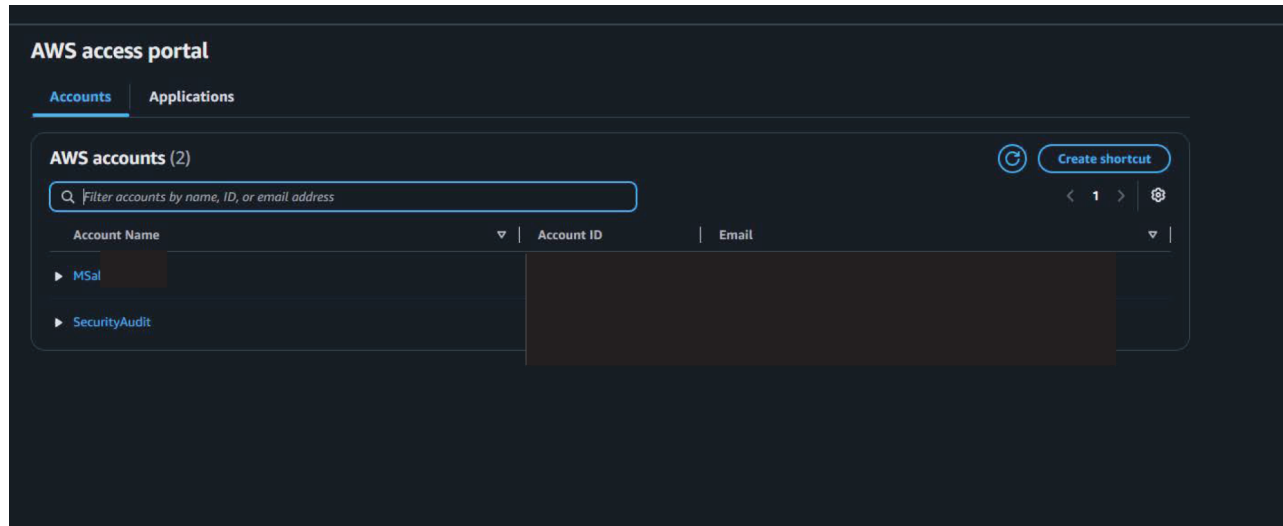
Creation time

[AdministratorAccess](#)

Full admin

2 minutes ago

The redacted review-and-submit evidence shows assignments to two AWS accounts. The selected users and groups section lists Administrators as a group, and the selected permission set is AdministratorAccess with description Full admin. Sensitive ARN and account values were redacted.

Figure 4 - SSO Access Portal Shows Assigned AWS Accounts

The redacted AWS access portal evidence shows two assigned AWS accounts available to the SSO user: the management account and SecurityAudit. Sensitive account IDs and email values were redacted.

Security Analysis

This project moves access away from individual IAM users and toward centralized SSO-based workforce access, which is more scalable for a multi-account organization.

Using the Administrators group means future administrative access can be managed by group membership and account assignments rather than by repeatedly creating direct user permissions.

AdministratorAccess is intentionally broad. It is useful for the lab's bootstrap phase, but a production environment should add lower-privilege permission sets for routine work.

IAM Identity Center provisions reserved roles inside target accounts behind the scenes. Those roles are assumed through the SSO flow rather than managed like normal hand-created IAM roles.

The AWS access portal evidence confirms the SSO user can see assigned accounts. A future hardening step should also capture CloudTrail events and verify session activity by SSO user identity.

Lessons Learned

A permission set is a collection of IAM policies that Identity Center can provision into target AWS accounts.

Access assignment in IAM Identity Center is a three-part relationship: user or group, permission set, and AWS account.

When a permission set is assigned to an account, AWS creates a reserved SSO role in that account and allows the federated Identity Center user to assume it.

The access portal is the user's normal entry point after SSO is configured and assigned. The regular IAM user path can be reserved for break-glass or special administrative cases.

Production Improvements

- Create separate least-privilege permission sets such as ReadOnlySecurityAudit, BillingReadOnly, and CloudTrailAdmin.
- Limit AdministratorAccess to break-glass or tightly controlled administrator groups.
- Use permission boundaries or custom permission sets for routine operations.
- Monitor Identity Center account assignments, permission set changes, and AWSReservedSSO role assumptions in CloudTrail.
- Require phishing-resistant MFA for privileged SSO groups where possible.
- Review assignments regularly and remove accounts or groups that no longer require administrative access.

Skills Demonstrated

IAM Identity Center	Permission sets	AWS Organizations
SSO account assignments	Group-based authorization	AdministratorAccess
AWS access portal	Federated role assumption	Multi-account administration

Resume Bullet

RESUME

Implemented AWS IAM Identity Center account assignments by creating an AdministratorAccess permission set, assigning it to the Administrators SSO group across two AWS accounts, and validating account availability through the AWS access portal.

STAR Interview Story

Situation: The IAM Identity Center foundation existed, but the SSO user still had no account-level authorization.

Task: Create an administrative permission set and assign it to the existing Administrators group across the AWS organization accounts.

Action: I created the AdministratorAccess permission set, selected the Administrators group, assigned it to two AWS accounts, submitted the provisioning workflow, and confirmed account visibility in the AWS access portal.

Result: The SSO user now has a centralized administrator access path through IAM Identity Center, completing the transition from setup-only SSO to functional multi-account access.

Version History

v1.0.0	July 13, 2026	Initial documented implementation. Created AdministratorAccess permission set, assigned it to the Administrators group across two AWS accounts, validated SSO access portal account visibility, documented evidence limitations, and embedded all provided project evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Another Me? SSO with IAM Identity Center, Part 2.
- User-provided AdministratorAccess permission set screenshot.
- User-provided account assignment review screenshot.
- User-provided AWS access portal confirmation screenshot.
- Project 13 - IAM Identity Center SSO and MFA Foundation.